

Homework for Chap 14

Figure 1 shows absorption spectra of GaAs bulk material close to the bandgap measured at different temperatures.

1. Using an equation, explain why the absorption show redshift with increasing temperature.
2. Remarkable absorption enhancement close the bandgap is observed at the temperatures $T = 21\text{ K}$, 90 K , and 186 K . However, no obvious absorption enhancement is observed at $T = 294\text{ K}$. Explain the reason.

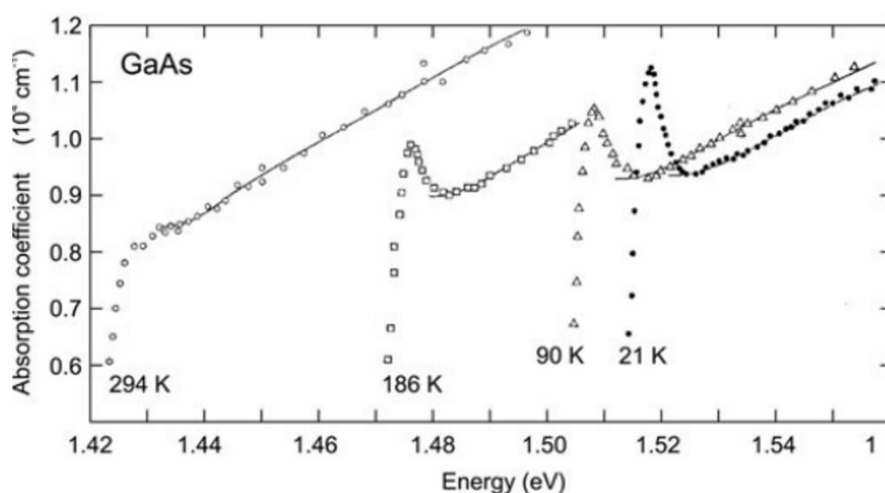


Figure 1. Absorption spectra of GaAs bulk material close to the bandgap at different temperatures.

$$1. \quad E_g(T) = E_g(0) - \frac{\alpha T^2}{T + \beta}$$

$$E_g(T)' = - \frac{2\alpha T(T + \beta) - \alpha T^2}{(T + \beta)^2} = - \frac{\alpha T^2 + 2\alpha \beta T}{(T + \beta)^2} < 0$$

随着 T 升, $E_g(T)$ 下降, 吸收能量下降, 红移

2. 吸收增加来源于激子, 但其在高温下不显著, 原因:

① 高温下, 激子束缚能相对于 $k_B T$ 较小
激子无法稳定存在

② T 升, 电子-声子散射增强, 导致吸收峰展宽, 使得
激子的特征被覆盖